

- C) A single phase bridge converter fed from a 250 V, 50 Hz supply is used to control the speed of a separately excited dc motor having an armature resistance of 1.5Ω and inductance 30 mH. The back emf at one speed of operation is 100 V and the converter control angle is 30° . Determine the average and rms values of the armature current, the power input to the motor and power factor of operation 6

Q.4 Solve Any Two of the following. 12

- A) Explain with neat diagram voltage source inverter fed Induction motor drive Understanding 6
- B) A slip ring induction motor incorporating slip energy recovery scheme for speed control drives a fan load having a characteristic of $T_L = KN^2$. The motor is rated at 440 V, 50 Hz, 100 kW. The speed is to be controlled from a rated value of 1420 to 750 rpm. The equivalent circuit parameters are $r_1 = 0.052 \text{ ohm}$, $r_2 = 0.06 \text{ ohm}$, $X_m = 10 \text{ ohm}$, $x_1 + x'_2 = 0.29 \text{ ohm}$, $\frac{T_p}{T_s} = 1.2$ Determine the firing angle range. Applying 6
- C) Explain in brief drives required for various stages in textile mills 6

Q. 5 Solve Any Two of the following. 12

- A) Explain in detail drives for cement mills application. Understanding 6
- B) Draw the block diagram of variable frequency control of multiple synchronous motor drive and explain in brief. Understanding 6
- C) Describe the static Kramer drive for speed control of 3-phase Slip ring IM. Draw and explain the speed torque characteristics of static Kramer drive? Understanding 6

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